

Course Name : Physics II Lab

Course Code : Phy 4242

Experiment No : 05

Experiment Name : Verification of Thevenin's
Theorem

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Objective:

- i) To be familiar with Thevenin's theorem and know its applicability.
- ii) To verify this theorem with the help of simple network.

Theory:

It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all other sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end. It states that, "Any two terminal linear bilateral network can be replaced by an equivalent circuit consisting of a voltage source, V_{th} and a series resistance, R_{th} ."

The value of V_{th} is the open circuit voltage across the terminals and R_{th} is the resistance measured

between the terminals with all the sources eliminated.
 The voltage sources are eliminated by shorting their terminals and the current sources are eliminated by opening their terminals.

Circuit Diagram :

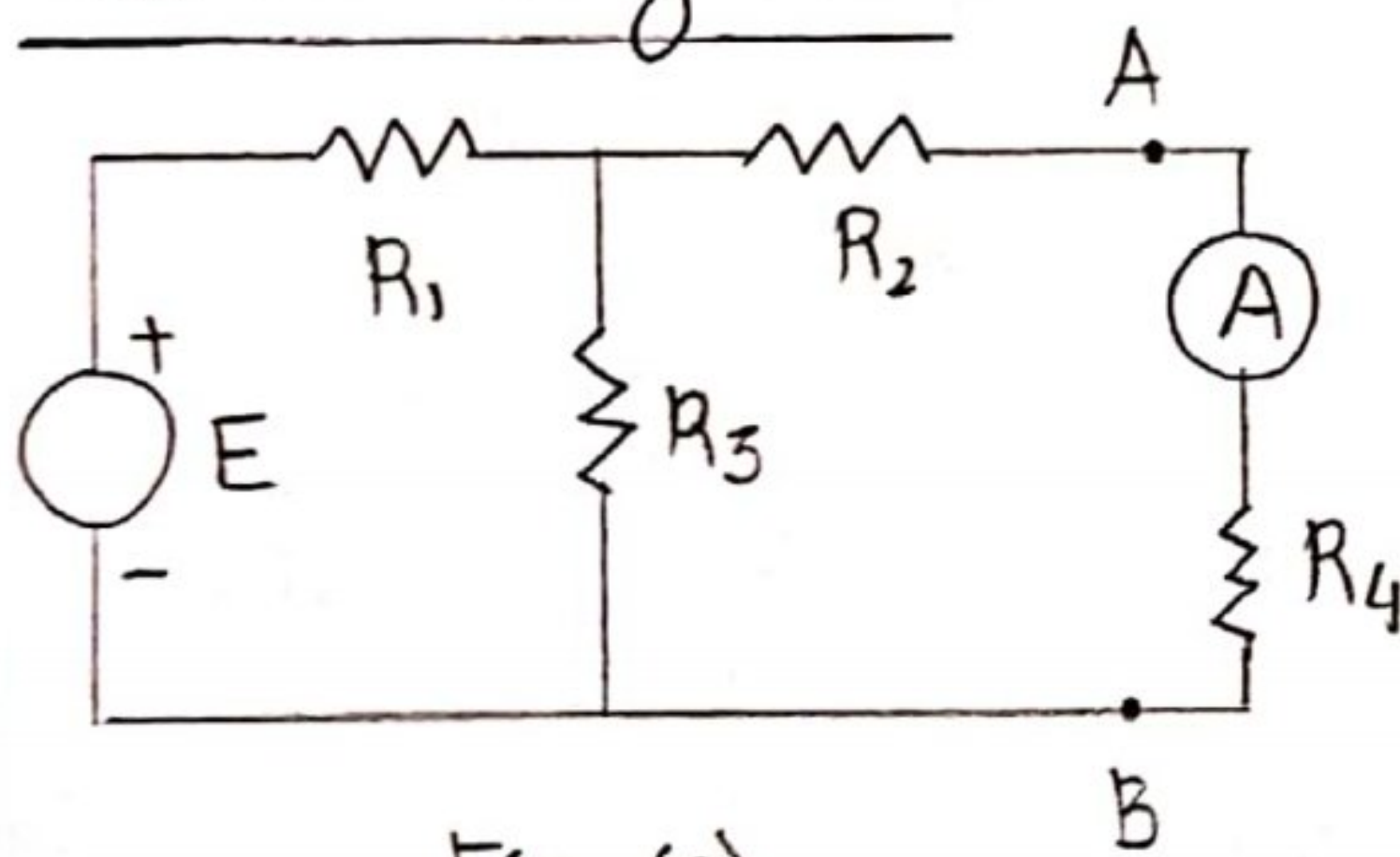


Fig. (a)

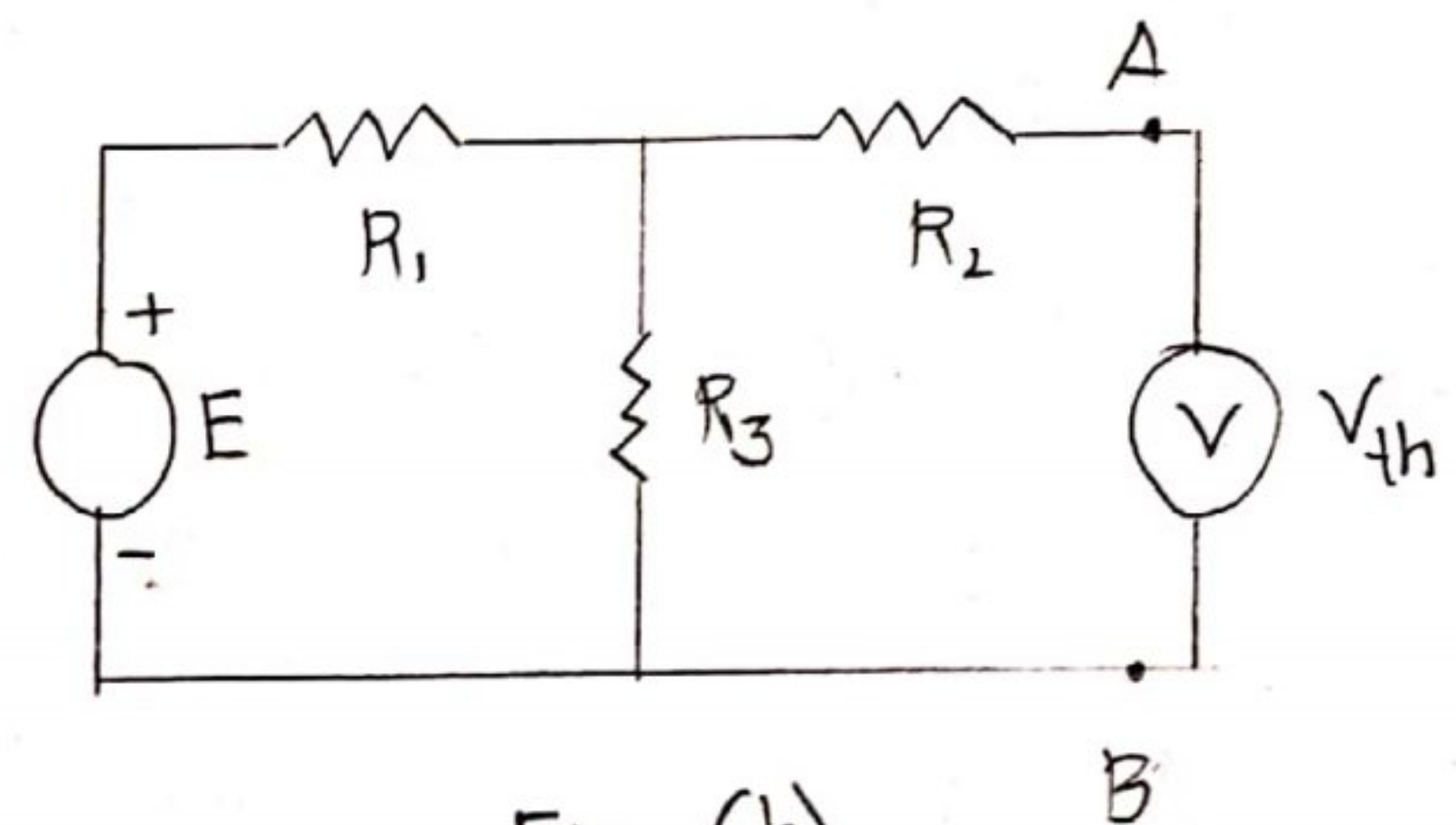


Fig. (b)

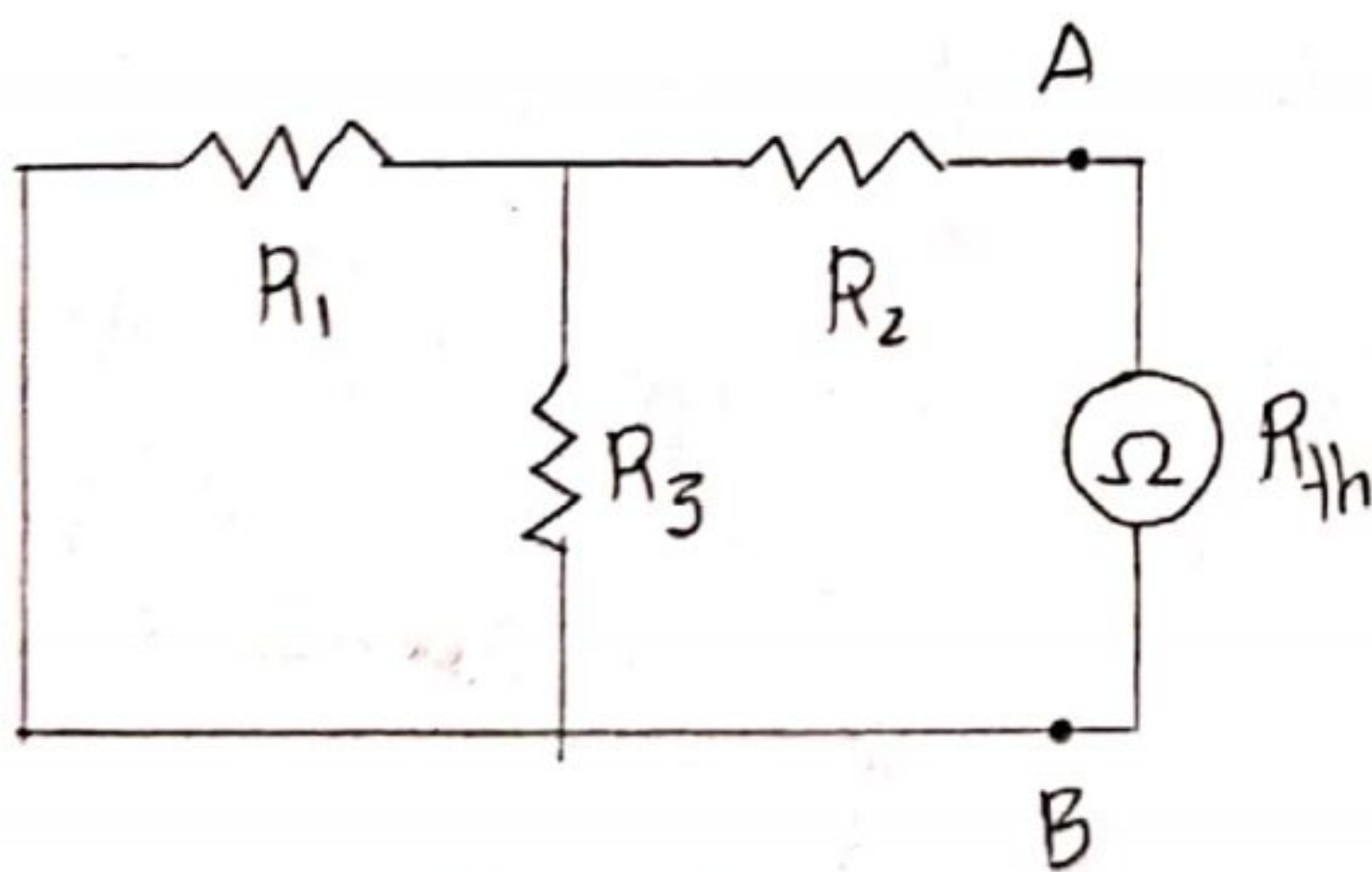


Fig. (c)

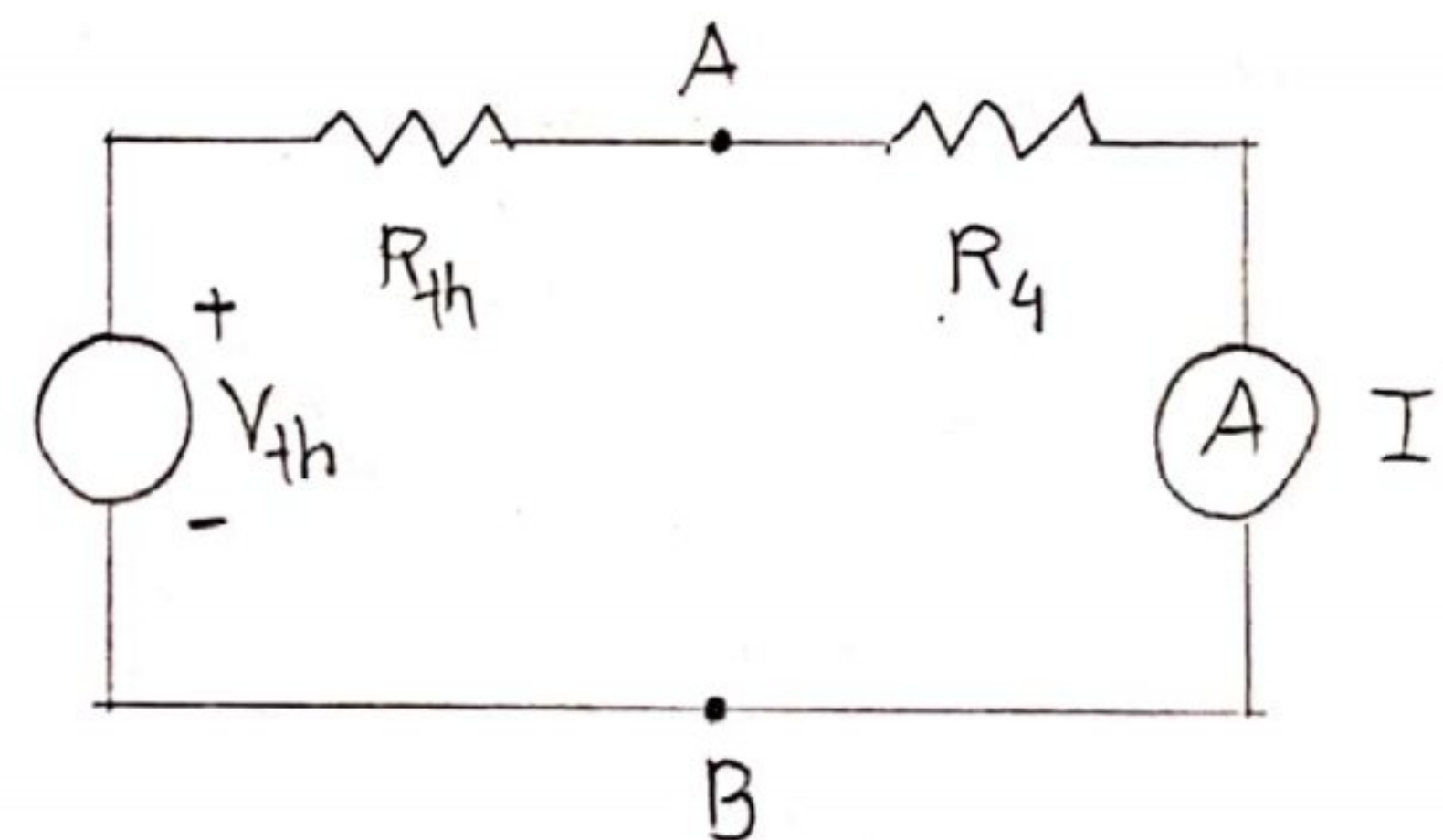


Fig (d)

Data Collection :

Observation no.	I (mA) Theoretical	V_{th} (V)	R_{th} (Ω)	I (mA) Practical
1	1.35	10.0	3.17k	1.40
2	1.35	4.76	3.135 k	1.25

Data Analysis:

Theoretical calculation,

For observation 1,

$$R_5 = R_2 + R_4$$

$$R_{eq} = (R_3 \parallel R_5) + R_1$$
$$= 3286 \Omega$$

$$I = \frac{R_5^{-1}}{R_5^{-1} + R_3^{-1}} \times 3.04 \text{ mA}$$
$$= 1.35 \text{ mA}$$

For observation 2,

$$R_{th} = (R_1 \parallel R_3) + R_2$$
$$= 3165 \Omega$$

$$V_{th} = \frac{2.11 \text{ k}}{2.11 \text{ k} + 2.11 \text{ k}} \times 10 = 5 \text{ V}$$

$$I = \frac{V_{th}}{R_{th} + R_4} = \frac{5}{3135 + 547}$$

$$= 1.35 \text{ mA}$$

Result:

We can see that the current through the terminals A and B are almost same. They are also close to the value obtained through theoretical calculation.

Discussion:

- i) While taking the readings, voltmeters should be connected in parallel and ammeters in series.
- ii) There should not be any loose connections.
- iii) Before starting the amount of resistance of the resistors were measured carefully.
- iv) Electricity shouldn't be flown in a circuit for a long time because the heat produced might affect the results.
- v) There are some slight discrepancies between the theoretical and practical values. This is primarily due to not being able to get the absolutely correct thevenin resistance with the supplied resistors.